

Paper List for Qualifying Exam

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- dp
 - [Calibrating Noise to Sensitivity in Private Data Analysis](#)
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 - [A Firm Foundation for Private Data Analysis](#)
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 - [Smooth Sensitivity and Sampling in Private Data Analysis](#)
Nissim K, Raskhodnikova S, Smith A.
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 - [Broadening the Scope of Differential Privacy Using Metrics](#)
Chatzikokolakis K, Andrés ME, Bordenabe NE, Palamidessi C.
PETS, 2013
 - [Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds](#)
Bun M, Steinke T.
TCC, 2016
- dp in Bayesian Inference
 - [Robust and Private Bayesian Inference](#)
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 - [On the Differential Privacy of Bayesian Inference](#)
Zhang Z, Rubinstein BI, Dimitrakakis C.
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- dp in adaptive data analysis
 - [Preserving Statistical Validity in Adaptive Data Analysis](#)
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STOC, 2015
 - [Generalization in Adaptive Data Analysis and Holdout Reuse](#)
Dwork C, Feldman V, Hardt M, Pitassi T, Reingold O, Roth A.
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 - [Algorithmic stability for adaptive data analysis](#)
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 - [Deep learning with differential privacy](#)
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- type system for dp
 - [Sensitivity Analysis Using Type-based Constraints](#)
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 - [Differential Privacy by Typing in Security Protocols](#)
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 - [Distance Makes the Types Grow Stronger: A calculus for differential privacy](#)
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 - [Linear Dependent Types for Differential Privacy](#)
Gaboardi M, Haeberlen A, Hsu J, Narayan A, Pierce BC.
POPL, 2013
 - [Higher-order approximate relational refinement types for mechanism design and differential privacy](#)
Barthe G, Gaboardi M, Gallego Arias EJ, Hsu J, Roth A, Strub PY.
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 - [Duet: An Expressive Higher-order Language and Linear Type System for Statically Enforcing Differential Privacy](#)
Near JP, Darais D, Abuah C, Stevens T, Gaddamadugu P, Wang L, Somani N, Zhang M, Sharma N, Shan A, Song D.
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- verification of privacy and utility
 - [The Foundational Cryptography Framework](#)
Petcher A, Morrisett G.
POST, 2015
 - [Proving differential privacy in hoare logic](#)
Barthe G, Gaboardi M, Arias EJ, Hsu J, Kunz C, Strub PY.
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 - [Proving Differential Privacy via Probabilistic Couplings](#)
Barthe G, Gaboardi M, Grégoire B, Hsu J, Strub PY.
LICS, 2016
 - [Verified Correctness and Security of mbedTLS HMAC-DRBG](#)
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 - [Formal certification of code-based cryptographic proofs](#)
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- dp with machine floating point
 - [Differential privacy with imperfect randomness](#)
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 - [On Significance of the Least Significant Bits For Differential Privacy](#)
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 - [Information Flow Inference for ML](#)
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 - [On Flow-Sensitive Security Types](#)
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 - [Dependent types in practical programming](#)
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 - [A Core Quantitative Coeffect Calculus](#)
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